

B.Sc. Mini Project

SIKSHA KENDRA

Online Learning & Quiz
Portal: Learn. Practice.
Evaluate.



Problem Statement

Content Fragmentation

Difficulty in providing organized subject content in one centralized location.

Accessibility Barriers

Students often lack quick, on-demand access to necessary learning materials.

Assessment Gaps

Challenges in conducting simple, immediate evaluations after a learning session.

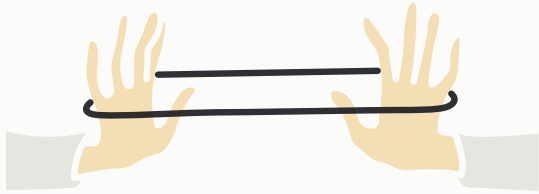
Manual Overhead

Time-consuming manual calculation of quiz results and performance tracking.

Maintenance Rigidity

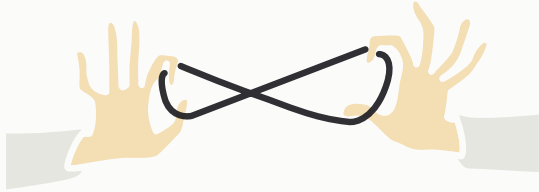
Updating content often requires modifying the entire application source code.

AI-Assisted Development Solution



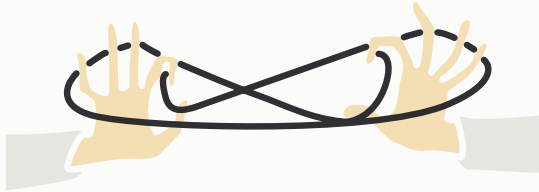
Manual Development

Slow, error-prone development process



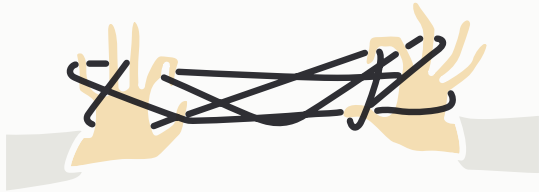
AI Generation

AI assists with structure, UI, and schemas



AI Logic Building

AI helps with complex quiz and authentication



Error Identification

AI identifies development errors early

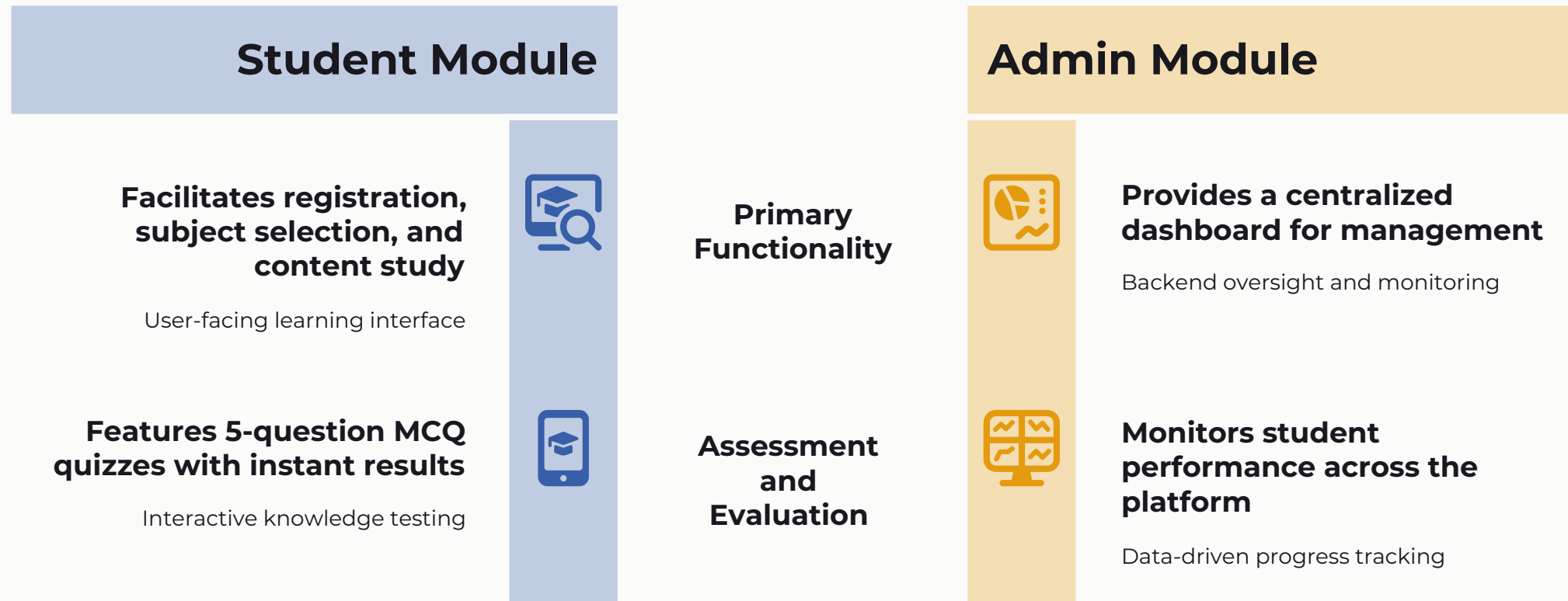


Accelerated Development

Faster, optimized, and error-free development

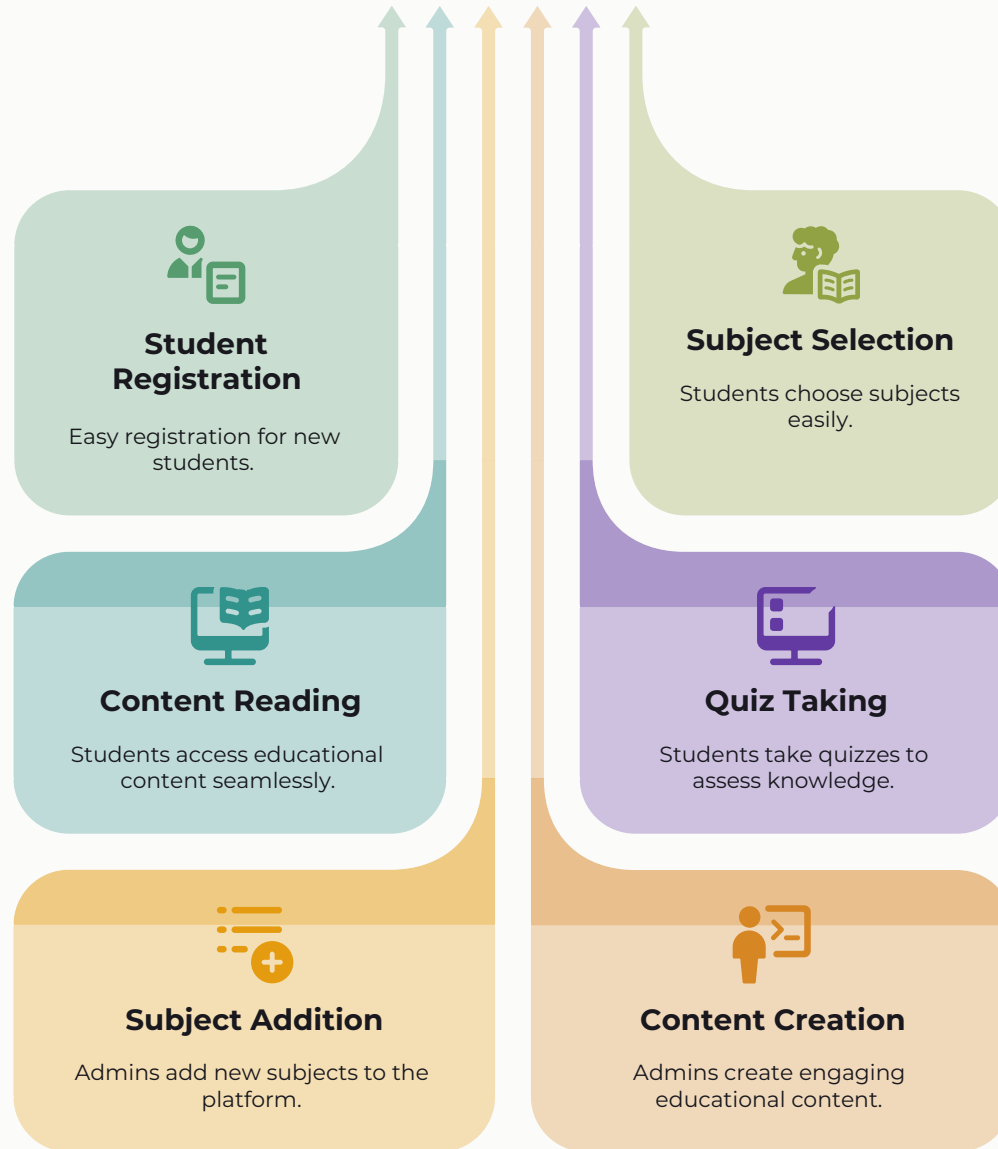
The project team maintains full control over final logic, content, and validation.

Which module of the Siksha Kendra system should you focus on?



How Siksha Kendra Works

Streamlined processes for both students and administrators.



Student Flow: Register → Login → Dashboard → Select Subject → Read → Quiz → Result. Admin Flow: Login → Dashboard → Add Subject → Add Content → Add MCQs → Publish.

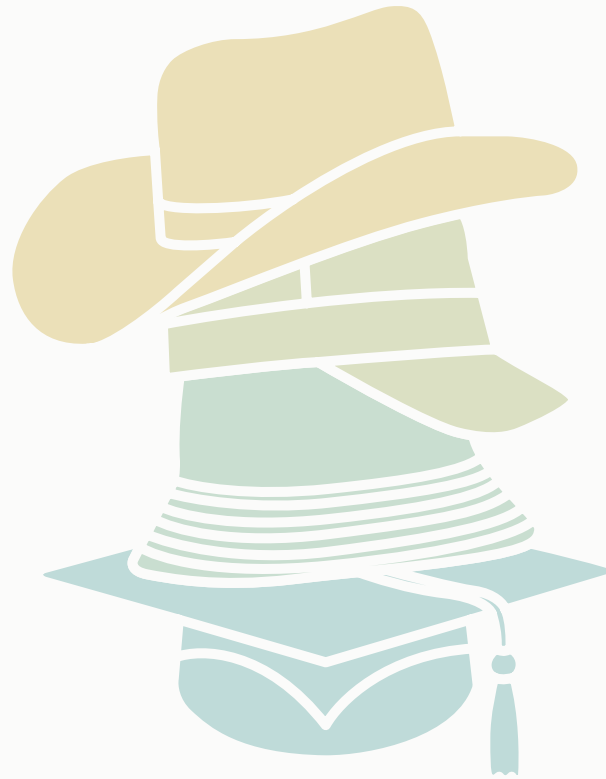
Technology Stack

Development

Lovable AI for assisted development and debugging

Frontend

React, TypeScript, and Tailwind
CSS for responsive UI



Security

Role-based access and RLS for data protection

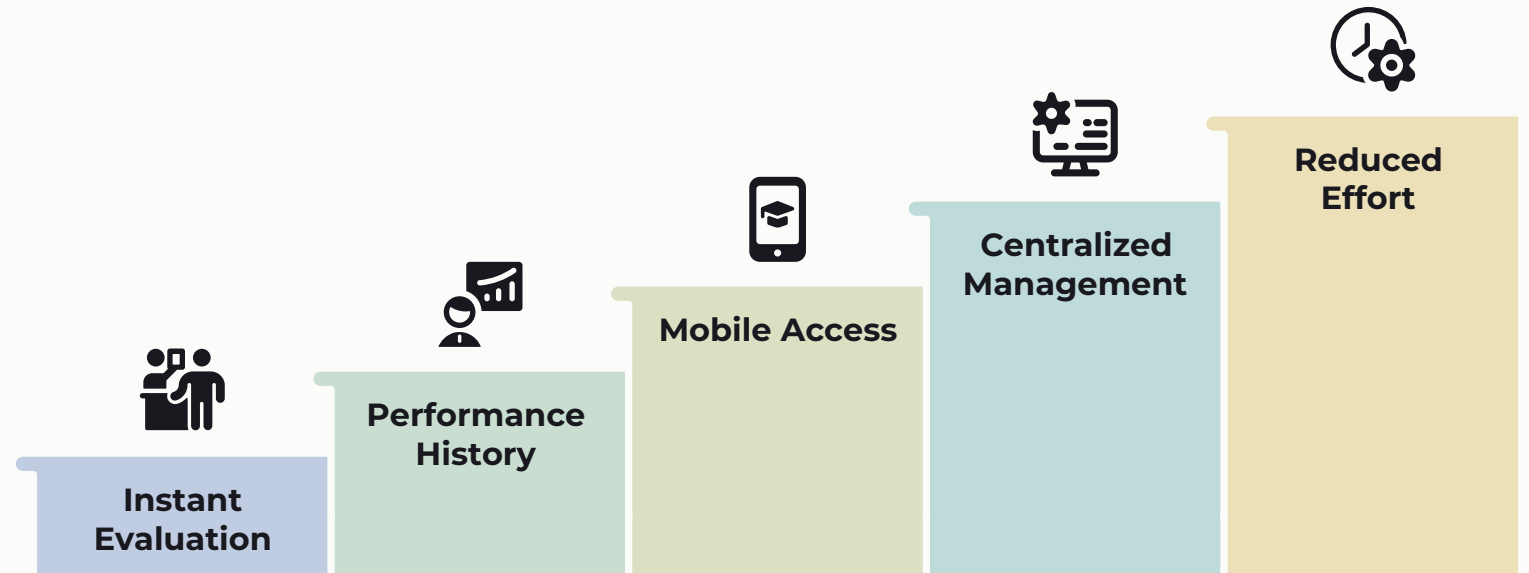
Backend

Supabase (PostgreSQL) for database and authentication

Key Features & Benefits

Enhancing the learning experience through automation and accessibility.

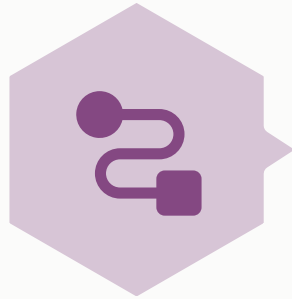
For Admins, centralized management allows content updates without touching source code, resulting in significant reduction in manual effort and time saved for both parties.



Pathways to Educational Growth

Development Roadmap

Larger question banks, timed exams, certificates, leaderboards, and advanced AI-assisted analytics.



Platform Adaptability

Adaptable for schools, colleges, coaching centers, and corporate training programs.

AI Question Generation

Automating the creation of quiz content.

Advanced Analytics

Deeper insights into student performance.

Personalized Learning

Tailoring content to individual student needs.

Mobile Application

Expanding reach through dedicated mobile platforms.

Advantages & Future Development

A solid foundation for a comprehensive learning platform.



Conclusion Thank You!

Siksha Kendra: Learn. Practice. Evaluate.

A practical, scalable foundation for digital learning and assessment.